

# Comparison of von Heijne and Support Vector Machine Models for Signal Peptide Prediction

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## Abstract

Signal peptides mediate protein translocation within the eukaryotic secretory pathway. Given the high costs associated with experimental validation, robust *in silico* prediction tools are of critical importance for applications ranging from genome annotation to disease diagnosis. This work compares the efficacy of von Heijne-based statistical methods against Support Vector Machines (SVMs) for signal peptide classification. Although the von Heijne method performs reliably on canonical motifs, it struggles with non-canonical sequence compositions. In contrast, the SVM model, trained on physicochemical features, achieved higher overall accuracy, proving to be a more effective tool for signal peptide prediction.

**Key words:** SP, SVM, ML, von Heijne

## Introduction

Intracellular protein distribution is a strictly regulated process, as the functional efficacy of a protein is contingent upon its delivery to the specific subcellular compartment that provides the requisite biochemical environment [1]. Numerous protein-sorting processes used by eukaryotic cells are mediated by N-terminal targeting signals [1]. Disruptions in these trafficking pathways or subsequent protein mislocalization are implicated in a broad spectrum of human pathologies [5].

Protein translocation into the eukaryotic secretory system or across prokaryotic membranes is coordinated by signal peptides (SPs), which are N-terminal sequences that are usually 15–30 residues long [17, 20].

As illustrated in Figure 1, SPs possess a characteristic tripartite architecture:

1. **The N-region:** An amino-terminal segment enriched with positively charged residues, such as lysine, arginine, and histidine.

2. **The H-region:** A central hydrophobic core, which frequently adopts an  $\alpha$ -helix conformation.
3. **The C-region:** A polar/neutral carboxyl-terminal segment containing a weakly conserved cleavage site that facilitates precise proteolytic processing following translocation [23].

SPs are essential to many biotechnological and medicinal applications, such as genome annotation, recombinant protein expression, diagnostic discovery, and vaccine creation, because of these structural characteristics [17, 20, 23].

Accurately identifying SPs is essential to predicting subcellular localization because they are the major factors for protein entrance into the secretory pathway. Nevertheless, experimental validation of SPs continues to be time-consuming, labor-intensive, and frequently vulnerable to cell-type or condition-specific variability [23, 6]. These constraints have catalyzed the development of *in silico* prediction tools, which provide scalable, rapid, and reproducible alternatives [13]. Computational inference of subcellular localization is now crucial for the functional annotation of uncharacterized proteomes by bridging the gap between raw sequence data and biological knowledge [8].

The methodological landscape for SP prediction has evolved significantly over several decades [16]. Early weight-matrix models [22] laid the groundwork for more sophisticated feature-based classifiers, such as Support Vector Machines (SVMs) [21, 7] and Random Forests [14]. Subsequent advancements introduced probabilistic and homology-based frameworks, including Hidden Markov Models [4], eventually culminating in modern deep-learning architectures [20, 18]. These deep-learning approaches currently define the state-of-the-art, as they can autonomously extract complex, non-linear sequence patterns from raw data and demonstrate superior generalization across diverse species and SP classes [20, 18, 2, 12].

The aim of this project is to evaluate the comparative efficacy of two distinct computational paradigms for SP detection: von Heijne's (vH) positional scoring matrices and an SVM-based machine-learning strategy. By benchmarking both methods against a common curated dataset, we assessed their



**Fig. 1.** Schematic representation of a canonical SP. The SP contains an N-terminal region (1-5 residues), an H-region (7-15 residues), and a C-region (3-7 residues)

capacity to capture the fundamental physicochemical properties of SPs and their ability to generalize to novel sequences. Our results indicate that while the vH method remains effective for identifying canonical motifs, the SVM approach achieves higher overall accuracy and Matthews Correlation Coefficient (MCC) by integrating diverse, sequence-derived features. Furthermore, our benchmarking highlights the characteristic false-positive and false-negative profiles inherent to each methodology, providing insights into their respective predictive limitations.

## Materials and Methods

### Computational Environment and Software Stack

The Python 3 programming environment was used to create the computational framework for this investigation, which adopted a modular architecture intended for repeatable bioinformatics analysis. The Pandas and NumPy libraries were used for data processing and structural manipulation of the main UniProtKB datasets. This allowed for effective numerical operations and the creation of high-dimensional feature vectors. To ensure environment stability and cross-platform reproducibility, the Conda package manager was utilized for dependency management and environment isolation. This allowed for a consistent execution context by precisely specifying library versions and managing the underlying system-level requirements.

The machine learning workflow was primarily managed via the Scikit-learn library. This included the entire pipeline from data preprocessing—specifically feature normalization using the StandardScaler—to the implementation of the Support Vector Machine (SVM) classifier utilizing the SVC module. Model performance was quantitatively rigorously assessed through metrics such as Accuracy and the Matthews Correlation Coefficient (MCC). Additionally, the classification efficacy was qualitatively evaluated through the generation of Precision-Recall (PR) curves.

For comparative benchmarking, a baseline was established using an implementation of the von Heijne (vH) algorithm [22], developed using NumPy for vectorized scoring matrix calculations. Statistical visualization and data exploration, including the generation of distribution plots and comparative performance charts, were executed using the Matplotlib and Seaborn libraries.

### Data Collection and Preprocessing

The UniProtKB database provided high-quality sequencing data for both signal peptide-positive (SP+) and signal peptide-negative (SP-) cases. A bespoke Python script created for reliable query management and data cleansing was used to automate data retrieval using the UniProt REST API. Several rigorous filters were used to create a high-confidence dataset of eukaryotic proteins with experimentally verified SPs. The first query required "Evidence at the protein level," was limited to "Reviewed" (Swiss-Prot) entries, and was restricted to the *Eukaryota* superkingdom. Additionally, fragments and sequences shorter than 40 residues were not included; only sequences with an empirically verified SP. As UniProtKB does not support direct filtering by SP length, an additional custom filter was implemented to retain only proteins possessing an SP of  $\geq 14$  residues. As of November 11, 2025, this process yielded an initial set of 2,956 entries, which was refined to 2,938 after length filtering.

**Table 1.** Composition of the benchmarking set and cross-validation training folds.

| Dataset Split | Negative    | Positive    | Total        |
|---------------|-------------|-------------|--------------|
| Benchmarking  | 1787        | 219         | 2006         |
| Fold 1        | 1429        | 175         | 1604         |
| Fold 2        | 1429        | 175         | 1604         |
| Fold 3        | 1429        | 175         | 1604         |
| Fold 4        | 1429        | 175         | 1604         |
| Fold 5        | 1430        | 175         | 1605         |
| <b>Total</b>  | <b>8933</b> | <b>1094</b> | <b>10027</b> |

The negative dataset was designed to represent non-secretory proteins with definitive subcellular localizations. The same quality filters (Eukaryota, reviewed, protein-level evidence, and minimum length) were applied. Critically, an exclusion filter was used to remove any protein with any reported evidence of an SP. To ensure a strictly non-secretory identity, proteins were required to have experimental evidence for localization in one of the following compartments: cytosol, nucleus, mitochondrion, plastid, peroxisome, or cell membrane. This query yielded 20,615 entries as of November 11, 2025.

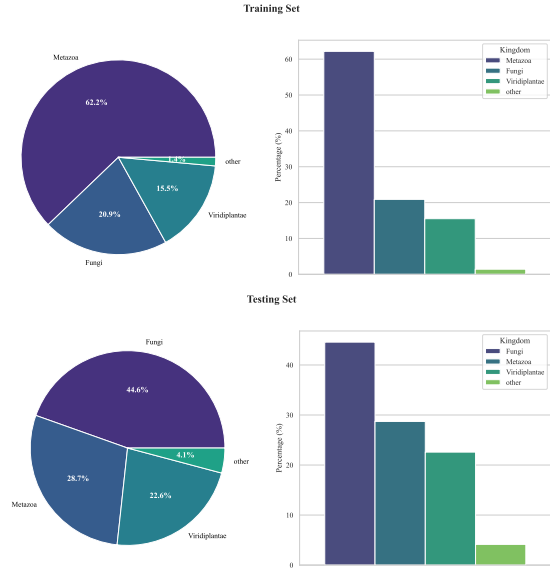
To mitigate biases arising from over-represented protein families or organisms, both datasets were subjected to a redundancy reduction pipeline. Sequences were clustered using MMseqs2 [19] with a minimum sequence identity threshold of 30%, an alignment coverage of 40%, and the connected component clustering mode. Selecting one representative sequence per cluster ensures a higher level of model generalization.

In order to preserve the initial positive-to-negative ratio, the generated non-redundant sequences were randomly divided into a training set (80%) and a benchmarking set (20%). The training data were further divided into five folds for cross-validation. Fold assignments were recorded in the metadata to ensure reproducibility and prevent data leakage. Associated metadata, including UniProt Accession, taxonomic lineage, SP cleavage site position, and the presence of transmembrane (TM) domains within the N-terminal 90 residues, were exported to a TSV file. For SP- sequences, the cleavage site was set to 0. For the TM domain feature, positive sequences were labeled "NIL," while negative sequences were assigned a Boolean flag. The final dataset statistics are summarized in Table 1.

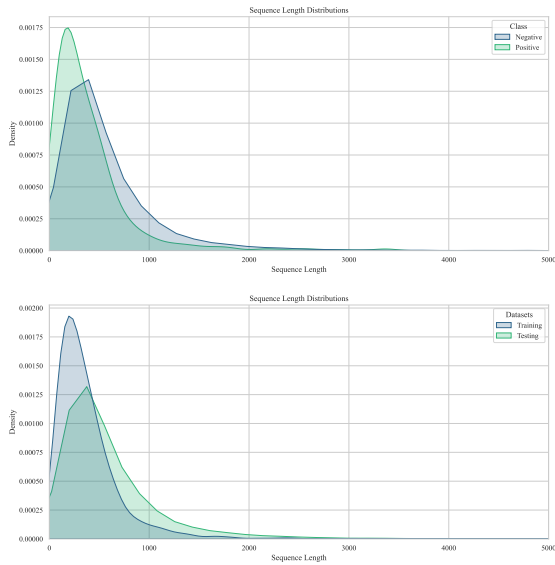
A descriptive analysis was conducted to verify that the training and benchmarking subsets were balanced and free of systematic bias. Both sets exhibited similar taxonomic distributions, with *Metazoa* as the most represented kingdom, followed by *Fungi* and *Viridiplantae*, and *Homo sapiens* as the dominant species. Protein length distributions were comparable, with most sequences falling within the 200–500 residue range.

In accordance with standard signal peptide profiles, SP lengths among SP+ sequences varied from roughly 15 to 60 amino acids, peaking at 20–25 residues. A high over-representation of apolar residues, especially leucine (which exhibits a two-fold enrichment) and alanine, together with a depletion of charged residues in comparison to the *UniProtKB/Swiss-Prot* baseline [3], were revealed by residue-composition analysis. Sequence-logo analysis, generated using WebLogo [11] for the region spanning 13 positions upstream to 2 positions downstream of the cleavage site, confirmed the

presence of the hydrophobic core and the highly conserved A-X-A motif across both subsets. Further details of these analyses are presented on figures (Figures 2–5).



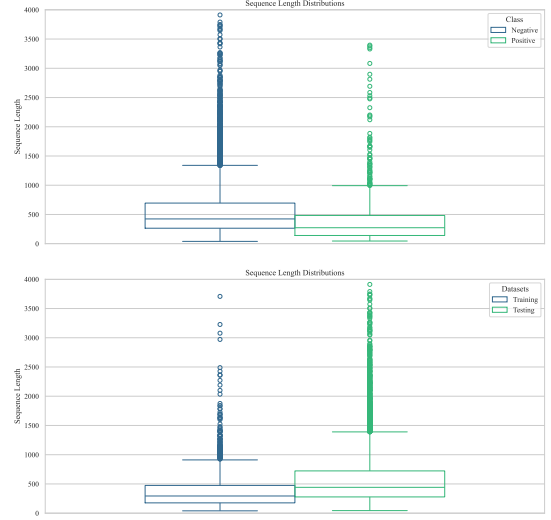
**Fig. 2.** Kingdom distribution across the datasets



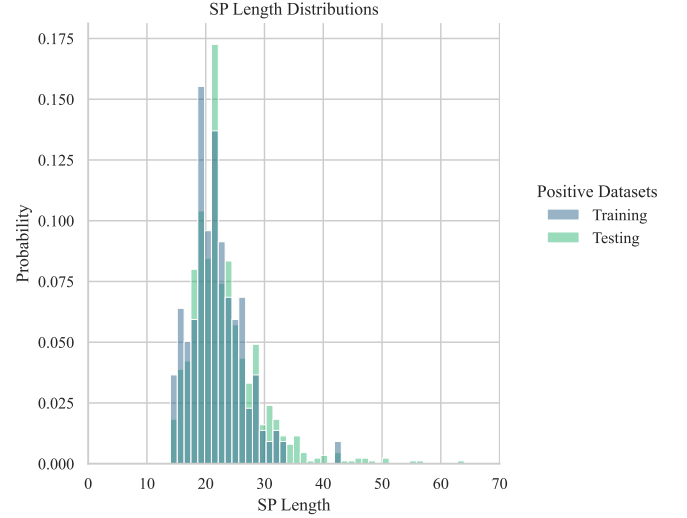
**Fig. 3.** Sequence length distribution: According to the class and according to the dataset

## von Heijne

A custom Python script was used to develop a classical weight matrix method based on the framework first proposed by von Heijne [22] in order to create a comparative baseline for signal peptide prediction. This method models the statistical distribution of amino acid residues within the SP motif using Position-Specific Weight Matrices (PSWMs).



**Fig. 4.** Sequence length distribution outliers: According to the class and according to the dataset



**Fig. 5.** SP length distribution

The model was trained on 15-residue sequence motifs extracted from the positive dataset. These motifs were anchored at the cleavage site, spanning 13 residues upstream and 2 residues downstream (positions  $-13$  to  $+2$ ), to capture the distinct signatures of the N-, H-, and C-regions. To construct the Position-Specific Probability Matrix (PSPM), residues were processed using a one-hot encoding strategy to count occurrences at each position  $j$ . To account for sparse data and prevent zero-probability values for residues not observed in the training set at specific positions, pseudocounts were incorporated. The probability  $M_{k,j}$  for residue  $k$  at position  $j$  was calculated as:

$$M_{k,j} = \frac{1 + \sum_{i=1}^N I(s_{i,j} = k)}{N + 20} \quad (1)$$

where  $N$  represents the total number of training sequences, the numerator includes a pseudocount of 1, and the

denominator is adjusted by the 20 standard amino acids. To determine the functional significance of each residue, the probability matrix was converted into a log-odds PSWM, normalized against the background distribution  $b_k$  derived from the *UniProtKB/Swiss-Prot* amino acid composition [3]:

$$W_{k,j} = \log_2 \left( \frac{M_{k,j}}{b_k} \right) \quad (2)$$

In this framework, positive weights indicate residues that are evolutionarily favored or functionally conserved at a given position, while negative values signify selective pressure against them.

To predict the presence of an SP in an uncharacterized query sequence  $X$ , a sliding window approach with a fixed length of 15 residues was employed. Given that SPs are localized to the N-terminus, the scanning process was restricted to the first 90 residues of each sequence. The score for a specific window was calculated by summing the corresponding weights from the PSWM (Figure 6):

$$\text{Score}(X|W) = \sum_{j=1}^L W_{x_j,j} \quad (3)$$

The final score for a sequence was defined as the maximum value obtained across all possible windows within the 90-residue N-terminal segment. This scoring mechanism was implemented using NumPy for optimized matrix operations. The optimal classification threshold was determined by maximizing the F1 score on the validation dataset during the cross-validation phase.

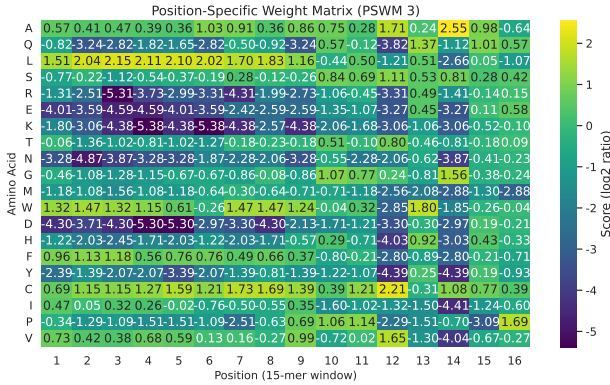


Fig. 6. The best PSWM derived from five cross-validation rounds

## Support Vector Machine (SVM)

Support Vector Machines (SVMs) are supervised learning algorithms that use an ideal hyperplane in a  $n$ -dimensional feature space to categorize data. This hyperplane is defined by the subset of training points that maximize the classification margin, known as support vectors. We used a "Soft Margin" formulation to deal with datasets that are not fully linearly separable. This formulation includes slack variables ( $\xi_i$ ) to accept degree-controlled misclassifications and outliers, improving the model's capacity to generalize to new biological sequences.

For non-linearly separable classification tasks, such as distinguishing signal peptide-carrying proteins from non-secretory proteins, the original feature space is mapped

into a higher-dimensional space where linear separation becomes feasible. To avoid the computational complexity of explicitly defining high-dimensional transformation functions, we employed the "kernel trick." This allows the direct evaluation of inner products in the transformed space using a kernel function,  $K(x_i, x_j)$ .

The decision boundary (hyperplane) in the feature space is defined by:

$$\langle w, x \rangle + b = 0 \quad (4)$$

where  $w$  is the weight vector orthogonal to the hyperplane and  $b$  is the bias term. The model is trained by solving the following constrained optimization problem:

$$\min_{w,b,\xi} \left( \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \right) \quad (5)$$

subject to:

$$y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \forall i \quad (6)$$

where  $C$  is the regularization parameter that balances margin maximization against training error minimization. The optimal solution is obtained via the Dual Lagrangian formulation:

$$\mathcal{L}(\alpha) = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j) \quad (7)$$

subject to  $0 \leq \alpha_i \leq C$  and  $\sum \alpha_i y_i = 0$ . For this study, the Radial Basis Function (RBF) kernel was selected to capture non-linear relationships between sequence features.

Protein sequences were transformed into 39-dimensional numerical feature vectors using a custom Python script. The feature set was divided into two main categories:

1. **Baseline Compositional Features (n=20):** These represent the global amino acid composition within the first 40 residues of the N-terminus. This captures the relative enrichment of specific residues, such as the high proportion of leucine and alanine characteristic of the SP hydrophobic core.
2. **Context-Dependent Physicochemical Features (n=19):** To account for local sequence properties, a sliding-window approach was implemented using several biological scales:
  - **Hydrophobicity and Aliphaticity (n=6):** Extracted using a window size of 6, based on the Kyte–Doolittle scale [15].
  - **Secondary Structure Propensity (n=6):**  $\alpha$ -helix and  $\beta$ -sheet propensities were calculated using a window size of 9, based on the Chou–Fasman scales [9, 10].
  - **Residue Charge (n=4):** Net charge distribution was assessed using a window size of 3.
  - **Transmembrane Propensity (n=3):** Evaluated with a window size of 7, based on the transmembrane tendency scale [24].

The SVM classifier was developed using a five-fold cross-validation procedure. Hyperparameter optimization for  $C$  and  $\gamma$  was performed via a systematic grid search on the training folds. To refine the feature set, a Random Forest (RF) classifier was concurrently employed to calculate Gini feature importance. The final SVM architecture was determined by evaluating the Matthews Correlation Coefficient (MCC) across progressively larger feature subsets, retaining only those that contributed significantly to discriminative performance and minimizing redundancy.

## Model Selection

### von Heijne

For the vH-based classifier, model selection focused on identifying the optimal decision threshold to maximize classification accuracy. During each cross-validation iteration, the PSWM was parameterized using 15-residue motifs anchored at the cleavage site of the positive training sequences, with background frequencies derived from the *UniProtKB/Swiss-Prot* database.

The 90-residue N-terminal fragments of sequences within the validation fold were scanned using the sliding window approach, with the maximum observed PSWM score assigned as the representative sequence-level score. To account for the inherent class imbalance between SP-positive and SP-negative examples, precision-recall (PR) curves were generated from these scores. The optimal decision threshold was identified as the value that maximized the  $F_1$ -score, ensuring a balanced trade-off between sensitivity and specificity. This procedure was repeated across all five cross-validation rounds; Among all the rounds, the model with the highest  $F_1$ -score was selected (Figure 6). This optimized threshold was subsequently evaluated on the independent testing fold to calculate performance metrics.

### SVM

Following feature encoding and scaling, the SVM classifier was optimized through a multi-stage process within a five-fold cross-validation framework. To minimize the risk of overfitting and ensure model robustness, an exhaustive grid search was initially performed over the  $C$  and  $\gamma$  hyperparameters.

The regularization parameter  $C$  was evaluated across a range of values (0.1, 1, 10, 100). This parameter governs the trade-off between maximizing the margin of the decision boundary and minimizing training classification errors; lower values of  $C$  promote a more generalized model by increasing tolerance to misclassifications, while higher values prioritize strict separation. Concurrently, the kernel coefficient  $\gamma$  was tested at values of "scale," 0.01, 0.1, and 1.0. While lower  $\gamma$  values prevent the model from becoming overly sensitive to individual data fluctuations, higher values allow the model to capture complex local variations, albeit with an increased risk of overfitting. For each  $C/\gamma$  combination, the model was trained on the training subsets and evaluated on the validation folds using the Matthews Correlation Coefficient (MCC).

To identify the most discriminative variables, a class-weighted Random Forest (RF) classifier was employed to calculate Gini feature importances. Features were subsequently ranked in descending order of their importance.

The top 1 to 30 features were created in incremental subgroups based on this ranking. Using the ideal hyperparameters, an SVM was trained for each subset. The ideal feature subset size that maximized predictive performance while preserving model parsimony was found by evaluating the MCC on the validation folds.

Once the optimal feature subset was identified, a secondary, refined grid search was conducted. The final model parameters and feature set were selected based on the peak MCC achieved across the validation sets. This two-stage optimization ensured that the final classifier was tuned specifically to the most informative physicochemical properties of the sequences.

## Performance Assessment

The performance of the binary classifiers in predicting signal peptides was evaluated using a comprehensive suite

of quantitative metrics. These measures are critical for assessing model efficacy, benchmarking different algorithms, and identifying potential classification biases. All performance indicators are derived from the confusion matrix, which cross-references the experimentally verified ground truth against the model's predictions.

The  $2 \times 2$  confusion matrix yields four fundamental outcomes:

- **True Positives (TP):** Secreted proteins correctly identified as SP-carrying.
- **True Negatives (TN):** Non-secretory proteins correctly identified as lacking an SP.
- **False Positives (FP):** Non-secretory proteins incorrectly classified as positive (Type I error).
- **False Negatives (FN):** Secreted proteins incorrectly classified as negative (Type II error).

To provide a balanced assessment, particularly in the context of the class imbalance inherent to biological sequence data, the following metrics were calculated:

### Matthews Correlation Coefficient (MCC)

The MCC is considered the most robust and informative metric for binary classification on imbalanced datasets. Unlike accuracy, the MCC only achieves a high score if the classifier performs well across all four quadrants of the confusion matrix. Its values range from  $-1$  (total disagreement) to  $+1$  (perfect prediction), with 0 representing random classification. It is defined as:

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (8)$$

### Accuracy (ACC)

The overall predictive power is quantified by Accuracy, which represents the proportion of total correct predictions relative to the entire dataset:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (9)$$

### Precision and Recall

To evaluate model performance specifically regarding the positive class, we employed Precision and Recall. Precision, or Positive Predictive Value (PPV), quantifies the reliability of positive predictions:

$$PPV = \frac{TP}{TP + FP} \quad (10)$$

Recall, also known as Sensitivity or the True Positive Rate (TPR), measures the model's ability to identify all existing positive instances (completeness):

$$TPR = \frac{TP}{TP + FN} \quad (11)$$

### $F_1$ Score

The  $F_1$  score represents the harmonic mean of Precision and Recall, providing a single balanced measure of the model's performance on the positive class:

$$F_1 = 2 \times \frac{PPV \times TPR}{PPV + TPR} \quad (12)$$

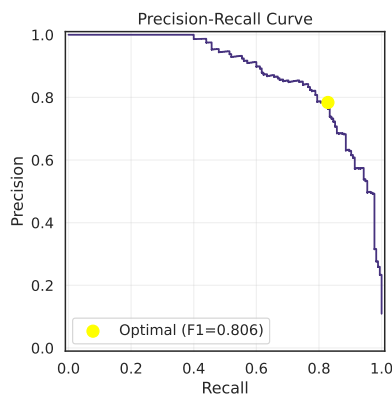
## Results

### Cross-Validation Performance

The predictive efficacy of the von Heijne (vH) and SVM models was first evaluated through a five-fold cross-validation (CV) procedure. For the vH method, an optimal classification threshold was determined for each fold by maximizing the  $F_1$  score on the validation set. The resulting thresholds and performance metrics are summarized in Table 2. The best optimal threshold across all folds was  $th = 8.159$ . Classification performance was further visualized via Precision–Recall (PR) curves (Figure 7).

**Table 2.** Per-fold optimal threshold values for the vH method.

| Fold | Optimal Threshold | F1 Score     |
|------|-------------------|--------------|
| 1    | 9.189             | 0.731        |
| 2    | 10.455            | 0.750        |
| 3    | <b>8.159</b>      | <b>0.806</b> |
| 4    | 8.250             | 0.646        |
| 5    | 8.001             | 0.708        |



**Fig. 7.** Precision-Recall Curve for the von Heijne classifier

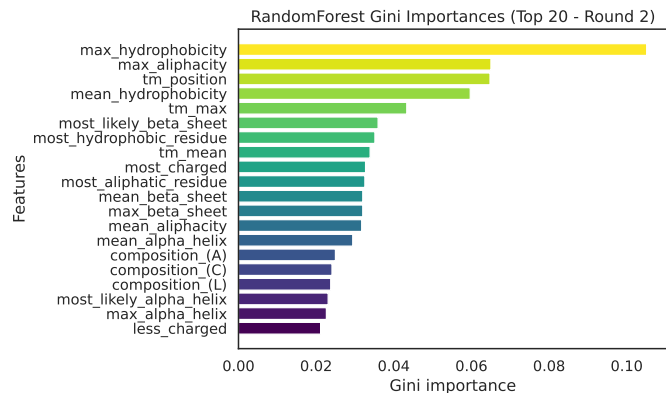
For the SVM model, the CV framework served to identify the optimal feature subsets by measuring MCC in each round. The top-20 features in the best round are illustrated (Figure 8). Then the model was trained using these features.

### Benchmarking on Hold-out Data

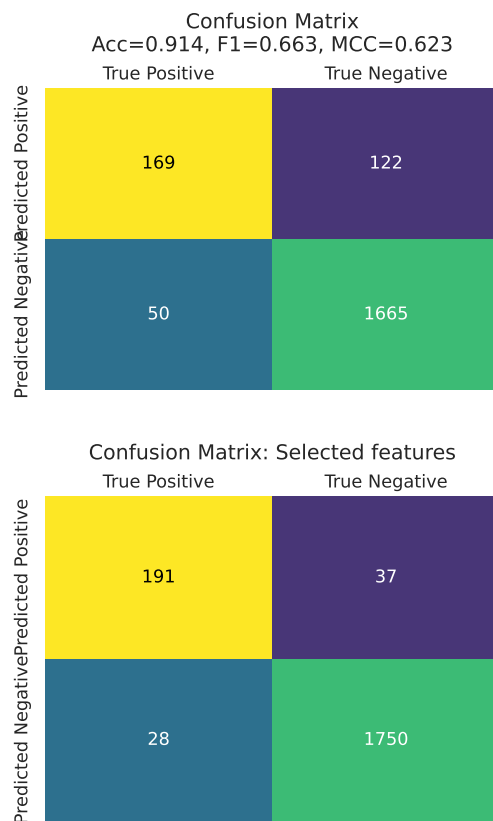
To evaluate final model performance, both classifiers were retrained on the complete training set and assessed against a separate benchmarking dataset. The benchmarking results (Table 3) confirm the trends observed during cross-validation. The raw classification counts further illustrate the SVM’s superior ability to minimize both False Positives and False Negatives (Figure 9).

**Table 3.** Benchmarking performance on hold-out dataset.

| Model | Acc.  | Prec. | Rec.  | $F_1$ | MCC   |
|-------|-------|-------|-------|-------|-------|
| vH    | 0.914 | 0.581 | 0.772 | 0.663 | 0.623 |
| SVM   | 0.964 | 0.862 | 0.837 | 0.850 | 0.829 |



**Fig. 8.** Top 20 training features chosen by CV framework



**Fig. 9.** Confusion Matrices: von Heijne and SVM

### Analysis of Misclassifications: FP and FN Profiles

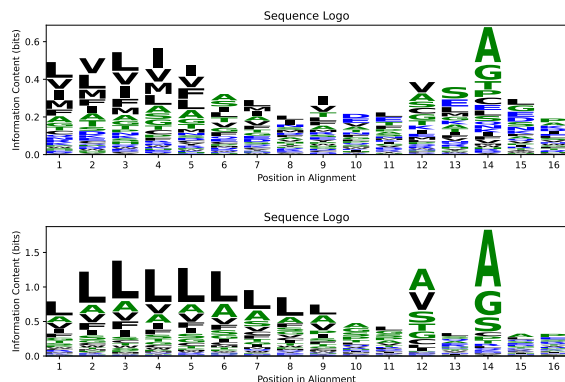
A detailed analysis of misclassified sequences was conducted to identify common sources of error. As shown in Table 4, the False Positive Rate (FPR) for TM-containing sequences was substantially higher than the overall FPR for both models.

For the vH method, the sequence logo was computed for all 15-residue fragments of the FNs in the training dataset. It is compared to the expected one in Figure 10, where it’s evident

**Table 4.** Overall vs. TM domain-related FPR.

| Model | Overall FPR | TM domain FPR |
|-------|-------------|---------------|
| vH    | 0.049       | 0.214         |
| SVM   | 0.020       | 0.070         |

that FNs may be due to a different composition of the cleavage-site context than expected.

**Fig. 10.** Sequence logos: FNs and expected

## Conclusions

Two different computational approaches for the prediction of signal peptides (SPs) in eukaryotic proteins were systematically evaluated in this work: a machine-learning framework based on Support Vector Machines (SVM) and the traditional von Heijne (vH) positional score matrix. We guaranteed that our results represent the genuine generalization capacity of the models rather than data artifacts by using a carefully selected dataset of empirically validated secretory and non-secretory proteins, which is controlled by stringent quality filters and strong train-test isolation.

Our analysis demonstrates that while the vH method effectively captures conserved sequence motifs adjacent to the cleavage site, its predictive efficacy is constrained by its reliance on local positional statistics and its limited sensitivity to global physicochemical trends. In contrast, the SVM classifier—incorporating amino acid composition, physicochemical properties, structural propensities, and context-dependent windowed features—exhibited significantly superior performance in both cross-validation and independent benchmarking. The marked improvements in MCC,  $F_1$  score, and Precision highlight the necessity of integrating diverse sequence-derived descriptors to successfully discriminate SPs from structurally similar motifs, such as N-terminal transmembrane segments. Feature importance analysis further substantiated that hydrophobicity patterns, secondary-structure propensities, and transmembrane tendency scores are critical discriminative features in modern SP prediction pipelines.

The detailed error profiling of both classifiers provided biologically meaningful insights into the challenges of protein-sorting prediction. False positives were predominantly enriched with proteins containing N-terminal transmembrane helices, reflecting the high degree of physicochemical mimicry between

these domains and the hydrophobic cores of SPs. Conversely, false negatives were typically associated with atypical SP compositions or non-canonical cleavage-site distributions. These results not only validate the internal logic of the models but also define specific subclasses of biologically divergent SPs that necessitate more expressive computational architectures.

In conclusion, machine-learning models utilizing high-dimensional feature spaces provide the precision and resilience needed for contemporary proteomics, even though traditional motif-based techniques are still useful as transparent, computationally light baselines. This comparative paradigm is essential for understanding how gradual improvements in model complexity result in quantifiable predictive benefits as deep-learning architectures continue to push the boundaries of the field. In the end, improving SP detection technologies boosts downstream applications in recombinant protein engineering and clinical variation interpretation, promotes large-scale genomics initiatives, and improves the accuracy of functional protein annotation.

## Acknowledgements

We wish to acknowledge Professors Castrense Savojardo and Matteo Manfredi for their expert advice, insightful feedback, and invaluable guidance throughout the completion of this work.

## Availability

All code, preprocessing scripts, trained models, and benchmarking datasets used in this study are openly available in the project's GitHub repository: [https://github.com/KinitaL/SP\\_predictors\\_evaluation](https://github.com/KinitaL/SP_predictors_evaluation). The repository provides the complete workflow for dataset assembly, feature extraction, von Heijne scoring, SVM training, cross-validation, and benchmarking, enabling full reproducibility of the analyses.

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